

BankOne Documentation

Living documentation for the BankOne core banking platform. **Docs are part of the source code** and must stay synchronized with every feature change.

Repository layout

Path	Description
BankOne-BackEnd/	Spring Boot 4 / Java 21 backend (WAR, Open Liberty)
BankOne-Frontend/	Angular frontend (<code>ng serve</code> → port 4200)
docs/	This documentation set

Quick start

Component	How to run	Default URL
Backend (Liberty)	<code>BankOne/scripts/redeploy-liberty.sh</code> (JDWP debug attach <code>:7777</code>)	<code>http://localhost:9080</code>
Kafka + Mailpit	<code>docker compose up -d kafka mailpit</code>	Kafka <code>:9092</code> / Mailpit UI <code>:8025</code>
Redis (local)	<code>docker compose up -d</code> (Redis service)	<code>:6379</code> — cache + rate limit
Load-test seed	<code>`psql ... -f scripts/seed-loadtest-10k.sql</code>	<code>` 10k LoadTest rows</code>
Backend (embedded Boot)	<code>mvn spring-boot:run</code>	<code>http://localhost:8080</code>
Frontend	<code>cd BankOne-Frontend && ng serve</code>	<code>http://localhost:4200</code>

Default staff login: `admin` / `Admin@123`

API base used by the UI: `BankOne-Frontend/src/app/core/config/api-config.ts`

Documentation map

Document	Purpose
ARCHITECTURE.md	System layers, packages, deployment
TECH_STACK.md	Technologies & versions (keep current)
FUNCTIONAL_SPECIFICATION.md	What the product does (implemented vs planned)
DEVELOPMENT_GUIDE.md	Local setup, conventions, teach-while-building
API_DOCUMENTATION.md	REST endpoints, roles, payloads
DATABASE_SCHEMA.md	Tables, sequences, ER notes
CALL_FLOW.md	Call hierarchies for major flows
CLASS_DIAGRAM.md	Mermaid class / package diagrams
SEQUENCE_DIAGRAMS.md	Mermaid sequence diagrams
CHANGE_IMPACT_GUIDE.md	What breaks when you change X
EXTENSION_GUIDE.md	How to add modules safely
CHANGELOG.md	Append-only change history
INTERVIEW_QA.md	Interview questions and answers from codebase
TECH_LEARNING_PLAN.md	Platform engineering learning roadmap (rate limit, cache, idempotency, sharding lab, ...)

Modules

Module	Status	Doc
Authentication	Implemented	MODULES/Authentication.md
Customer	Implemented	MODULES/Customer.md
Account	Implemented	MODULES/Account.md
Notification	Implemented	MODULES/Notification.md (Kafka → email)
Employee	Implemented	MODULES/Employee.md (multi-role staff)
Transaction	Implemented	MODULES/Transaction.md (ledger + staff list UI)
Portal & approvals	Implemented	MODULES/Portal.md (beneficiaries, pending transfers)
Reports	Implemented	MODULES/Reports.md (charts + PDF)
Audit	Implemented	MODULES/Audit.md (event trail + backfill)
Caching (Redis)	Implemented (local)	MODULES/Caching.md (cache-aside; prod off)
Branch	Not implemented (branch code string only)	MODULES/Branch.md
Loan	Partial (<code>AccountType.LOAN</code> only)	MODULES/Loan.md

Documentation ownership rule

Whenever code changes, update the affected docs **in the same task**, then append `CHANGELOG.md`. See `.cursor/rules/documentation-architect.mdc`.